

Computational report (Graphene and carbon nanotubes)

Stef Gaudenzio Lootens

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1 Brief review of the theoretical background

1.1 Crystal structure, real and reciprocal space

Crystals are solid materials characterized by their translational symmetries. More formally, we can identify simple crystals by a lattice that describes the periodic arrangement of atoms. A Bravais lattice is defined by 3 lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ (or just 2 in 2-dimensional space), often also called primitive lattice vectors. All possible lattice translation vectors that describe the positions of the crystal atoms can then be obtained by:

$$\mathbf{T} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

Where n_1, n_2 and n_3 are integers. For each crystal we can also introduce the reciprocal (primitive) lattice vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ that have the dimensions of wave vectors and describe the reciprocal space symmetry. They are defined by the following property:

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}$$

For a 3D bravais lattice, they can be easily calculated with $\mathbf{G}_1 = \frac{2\pi}{\Omega} \mathbf{a}_2 \times \mathbf{a}_3$ and equivalent equations with cyclical permutations of the indices, where Ω is the volume of the real space primitive cell. This construction is useful for various reasons, for example any function that has the periodicity of the lattice can be written in a discrete Fourier expression in terms of reciprocal lattice vectors:

$$u(\mathbf{r} + \mathbf{T}) = u(\mathbf{r}) \quad -> \quad u(\mathbf{r}) = \sum_{\mathbf{G}_m} C_{\mathbf{G}_m} e^{i\mathbf{G}_m \cdot \mathbf{r}}$$

Where $\mathbf{G}_m = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + m_3 \mathbf{b}_3$ are the reciprocal lattice translation vectors. This property is a consequence of the fact that $\mathbf{T} \cdot \mathbf{G}_m = 2\pi k$ with k an integer.

Up to now we have constructed an ideal simple crystal. There also exist more complicated crystals that are not Bravais lattices but can be described in terms of a basis of atoms in addition to a Bravais lattice. More importantly, real crystals have finite dimensions so their periodicity doesn't repeat endlessly like a theoretical lattice. In many computational methods, this problem is solved by introducing the Born-Von Karman boundary conditions: $\Psi(\mathbf{r} + N_i \mathbf{a}_i) = \Psi(\mathbf{r})$ where i is each lattice dimension. Practically this means that the wave function itself is periodic and repeats itself after N_i unit cells and implies that wavefunctions will have discretized wave vectors in reciprocal space $\mathbf{q} = \sum_i \frac{k_i}{N_i} \mathbf{b}_i$ where k_i are integers. We also define "first Brillouin zone" in reciprocal space the group of points that are closer to the origin (Γ point) than any other reciprocal lattice point, its utility will become more clear later.

1.2 Bloch's theorem and the solution of the Schroedinger equation in a periodic potential.

Bloch's theorem is an important result that gives information on the kind of wavefunctions that are solutions of the Schroedinger equation in a periodic potential. It also allows to rewrite the problem in matrix form so that it becomes an eigenvalue problem, computationally solvable with the appropriate approximations. In practice, Bloch's theorem can be used to exploit the periodicity of a crystal to reduce the solution of the Schroedinger equation for an ideal solid to a single unit cell with a series of different boundary conditions.

Bloch's theorem statement: The solutions of the Schroedinger equation in a periodic potential are products of planewaves with functions that have the same periodicity of the potential, that is to say $\Psi_{n\mathbf{q}} = e^{i\mathbf{q}\mathbf{r}} u_{n\mathbf{q}}$ where $\mathbf{q}$ is a reciprocal space vector. Another way to state the theorem is that each solution has the same shape inside different lattice cells located at $\mathbf{R}_i$ and $\mathbf{R}_j$ besides a phase factor $e^{i\mathbf{q}(\mathbf{R}_i - \mathbf{R}_j)}$

Proof:

The time independent Schroedinger's equation states:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \Psi = E \Psi \quad (1)$$

With a series of clever manipulations it can be rewritten in reciprocal space in such a way that it directly implies the Bloch's theorem statement. Let's start by decomposing each function in planewave basis, which is a complete and discrete set of basis states:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \sum_{\mathbf{K}} V_{\mathbf{K}} e^{i\mathbf{K}\mathbf{r}} \right] \sum_{\mathbf{q}} c_{\mathbf{q}} e^{i\mathbf{q}\mathbf{r}} = E \sum_{\mathbf{q}} c_{\mathbf{q}} e^{i\mathbf{q}\mathbf{r}} \quad (2)$$

To respect the periodicity of V , $\mathbf{K}$ is a reciprocal lattice vector, while $\mathbf{q}$ is a vector in reciprocal space that satisfies the Born-Von Karman boundary conditions. Rearranging the terms and applying the laplacian we get:

$$\left[\frac{\hbar^2}{2m} q^2 - E \right] \sum_{\mathbf{q}} c_{\mathbf{q}} e^{i\mathbf{q}\mathbf{r}} + \sum_{\mathbf{K}} \sum_{\mathbf{q}} V_{\mathbf{K}} c_{\mathbf{q}} e^{i(\mathbf{q}+\mathbf{K})\mathbf{r}} = 0 \quad (3)$$

The second term can be rewritten by setting $\mathbf{q}' = \mathbf{q} + \mathbf{K}$, and then recasting $\mathbf{q}'$ and $\mathbf{K}$ into $\mathbf{q}$ and $\mathbf{K}'$ (since they are dummy indices):

$$\sum_{\mathbf{K}} \sum_{\mathbf{q}} V_{\mathbf{K}} c_{\mathbf{q}} e^{i(\mathbf{q}+\mathbf{K})\mathbf{r}} = \sum_{\mathbf{K}} \sum_{\mathbf{q}'} V_{\mathbf{K}} c_{\mathbf{q}'-\mathbf{K}} e^{i(\mathbf{q}')\mathbf{r}} = \sum_{\mathbf{K}'} \sum_{\mathbf{q}} V_{\mathbf{K}'} c_{\mathbf{q}-\mathbf{K}'} e^{i(\mathbf{q})\mathbf{r}} \quad (4)$$

The equation then becomes:

$$\sum_{\mathbf{q}} e^{i\mathbf{q}\mathbf{r}} \left[\left(\frac{\hbar^2}{2m} q^2 - E \right) c_{\mathbf{q}} + \sum_{\mathbf{K}'} V_{\mathbf{K}'} c_{\mathbf{q}-\mathbf{K}'} \right] = 0 \quad (5)$$

Since the planewaves $e^{i\mathbf{q}\mathbf{r}}$ form an ortogonal set, each term must vanish individually. Furthermore, it's convenient to recast each vector $\mathbf{q}$ into $\mathbf{k}-\mathbf{K}$, where $\mathbf{K}$ is a reciprocal lattice vector and $\mathbf{k}$ is inside the first Brillouin zone. After making this substitution, each term becomes:

$$\left[\frac{\hbar^2}{2m} (\mathbf{k}-\mathbf{K})^2 - E \right] c_{\mathbf{k}-\mathbf{K}} + \sum_{\mathbf{K}'} V_{\mathbf{K}'} c_{\mathbf{k}-\mathbf{K}'} = 0 \quad (6)$$

After making the substitution $\mathbf{K}'' = \mathbf{K}' + \mathbf{K}$ and then recasting $\mathbf{K}''$ into $\mathbf{K}'$ (Dummy index):

$$\left[\frac{\hbar^2}{2m} (\mathbf{k} - \mathbf{K})^2 - E \right] c_{\mathbf{k}-\mathbf{K}} + \sum_{\mathbf{K}'} V_{\mathbf{K}'-\mathbf{K}} c_{\mathbf{k}-\mathbf{K}'} = 0 \quad \forall \mathbf{k}, \mathbf{K}, \mathbf{K}' \quad (7)$$

Which is the final equation we wanted to obtain. This equation is valid for each $\mathbf{k}$ inside the first Brillouin zone and is an eigenvalue problem that only couples terms of the form $\mathbf{k} + \mathbf{K}$ with their translations by reciprocal lattice vectors $\mathbf{k} + \mathbf{K}'$, it doesn't couple different $\mathbf{k}$ and $\mathbf{k}'$ inside the first Brillouin zone: This is important because it allows to solve the problem independently for each $\mathbf{k}$.

If we visualize the problem in matrix form, it would be a block matrix with a different block for each $\mathbf{k}$, with the kinetic energy terms present on the diagonal. To be able to computationally solve this problem, we have to make some approximations. The first one is to take a finite number of $\mathbf{K}$'s, following the supposition that since functions with larger $\mathbf{K}$'s have higher energy, they will be less and less present/occupied. Below an explicit example of a single block of the matrix recasting the ensemble of equations, in the case of considering just 3 $\mathbf{K}$ vectors:

$$\begin{bmatrix} \left[\frac{\hbar^2}{2m} (\mathbf{k} - \mathbf{K}_1)^2 + V_0 \right] & V_{\mathbf{K}_2 - \mathbf{K}_1} & V_{\mathbf{K}_3 - \mathbf{K}_1} \\ V_{\mathbf{K}_1 - \mathbf{K}_2} & \left[\frac{\hbar^2}{2m} (\mathbf{k} - \mathbf{K}_2)^2 + V_0 \right] & V_{\mathbf{K}_3 - \mathbf{K}_2} \\ V_{\mathbf{K}_1 - \mathbf{K}_3} & V_{\mathbf{K}_2 - \mathbf{K}_3} & \left[\frac{\hbar^2}{2m} (\mathbf{k} - \mathbf{K}_3)^2 + V_0 \right] \end{bmatrix} \begin{bmatrix} c_{\mathbf{k}-\mathbf{K}_1} \\ c_{\mathbf{k}-\mathbf{K}_2} \\ c_{\mathbf{k}-\mathbf{K}_3} \end{bmatrix} = E \begin{bmatrix} c_{\mathbf{k}-\mathbf{K}_1} \\ c_{\mathbf{k}-\mathbf{K}_2} \\ c_{\mathbf{k}-\mathbf{K}_3} \end{bmatrix}$$

For each $\mathbf{k}$ we have thus an eigenvalue problem of the type $\mathbf{H}\mathbf{C} = E\mathbf{C}$, where $\mathbf{H}$ is the hamiltonian recast in matrix form, $\mathbf{C}$ the vector of coefficients and E the associated energy. By solving these eigenvalue problems, for each $\mathbf{k}$ we obtain the respective eigenenergies $E_{n\mathbf{k}}$ and the values of the indices $c_{\mathbf{k}-\mathbf{K}}$, that by construction identify the solutions to be compositions of the following functions:

$$\psi_{n\mathbf{k}} = \sum_{\mathbf{K}} c_{\mathbf{k}-\mathbf{K}} e^{i(\mathbf{k}-\mathbf{K})\mathbf{r}} = e^{i\mathbf{k}\mathbf{r}} \sum_{\mathbf{K}} c_{\mathbf{k}-\mathbf{K}} e^{-i\mathbf{K}\mathbf{r}} = e^{i\mathbf{k}\mathbf{r}} u_{n\mathbf{k}} \quad (8)$$

Where $u_{n\mathbf{k}}$ has the periodicity of the lattice, thus proving the Bloch theorem. The sets of eigenenergies $E_{n\mathbf{k}}$ for each $\mathbf{k}$ are discrete and can be grouped into *energy bands* (same n , different $\mathbf{k}$).

1.3 The Tight-binding approximation

The tight-binding method aims to reduce the difficulty of the band structure problem by reframing it in terms of a composition of systems we can more easily describe and understand, which are electrons in single atom orbitals: it defines the electronic structure problem in terms of localized orbitals (compared to the plane wave basis we have seen before). With respect to the real system it becomes an approximation that fits when the states are tightly bond to the nuclei but it works well in various cases and it has led to many computational applications.

The method essentially follows a linear combination of atomic orbitals (LCAO) approach, in which the atomic states are used as basis orbitals. These states are assumed to be available from some kind of electronic structure calculation and we can label them $u_p(\mathbf{r}-\mathbf{R})$ where the index p labels the levels of an atom located at $\mathbf{R}$. From these states, the Bloch basis functions are built in the following way:

$$\phi_{p,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} u_p(\mathbf{r}-\mathbf{R}) \quad (9)$$

And a general Bloch state is a linear combination of these:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \sum_p C_p(\mathbf{k}) \phi_{p,\mathbf{k}}(\mathbf{r}) \quad (10)$$

Note how these correspond to the same concept of $\psi_{n\mathbf{k}}$ in eq. [8], just constructed in a different way: it can be demonstrated that they satisfy the Bloch theorem condition $\phi_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}}\phi_{\mathbf{k}}(\mathbf{r})$. Since $u_p(\mathbf{r}-\mathbf{R})$ are given, these states are defined once we know $C_p(\mathbf{k})$. We know that these coefficients can be determined by solving the eigenvalue problem, in this case it becomes a generalized version because the basis functions are not an orthogonal set:

$$\mathbf{H}\mathbf{C}(\mathbf{k}) = E\mathbf{S}\mathbf{C}(\mathbf{k}) \quad (11)$$

where the vector $\mathbf{C}(\mathbf{k})$ has components $C_p(\mathbf{k})$, the coefficients that we want to determine. Note how this equation is similar to eq. [7] in the Bloch theorem demonstration, but in this case the basis functions have a different form and in general a non diagonal overlap. $\mathbf{H}$ and $\mathbf{S}$ are respectively the Hamiltonian and overlap matrices and their elements are given by:

$$H_{pq} = \langle \phi_{p,\mathbf{k}} | H | \phi_{q,\mathbf{k}} \rangle \quad (12)$$

$$S_{pq} = \langle \phi_{p,\mathbf{k}} | \phi_{q,\mathbf{k}} \rangle \quad (13)$$

Knowing the form of the basis functions from eq. [9] and using the periodicity of the lattice, we obtain the following expressions for the matrix elements:

$$H_{pq} = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \int d^3r u_p(\mathbf{r}-\mathbf{R}) H u_q(\mathbf{r}) \quad (14)$$

$$S_{pq} = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \int d^3r u_p(\mathbf{r}-\mathbf{R}) u_q(\mathbf{r}) \quad (15)$$

As the states u_p and u_q are rather strongly localised near the nuclei, they will have virtually no overlap when centred on atoms lying far apart. This allows to restrict the sums in [14] and [15] to only the first few shells of neighbouring atoms, sometimes only nearest neighbours. This reasonable simplification represents a strength of the tight-binding approximation; other strengths of this method are the fact that it's possible to use other types of basis functions not restricted to atomic orbitals that better suit specific systems, as long as they are localised, and the fact that some parameters (for example the overlap energies) can be taken from other types of simulations or experimental results instead of having to be calculated within this method.

1.4 Tight-binding calculation for graphene

We now consider and describe two possible applications of the tight-binding method, starting with graphene. Graphene is a 2D material made of just carbon atoms ordered into an hexagonal lattice. This sheet of atoms can be described by a 2D Bravais lattice of triangular shape with a basis of two atoms in the unit cell.

There are various possible choices for primitive vectors, for our purposes let's consider this example: $\mathbf{a}_1 = a(\sqrt{3}/2, 1/2)$, $\mathbf{a}_2 = a(\sqrt{3}/2, -1/2)$ (where $a = 2.461\text{\AA}$ for the real material) with the two atoms, that we can distinguish and label A and B, in $(0, 0)$ and $(2/\sqrt{3}, 0)$ in cartesian coordinates. With this choice, the corresponding reciprocal lattice vectors become $\mathbf{b}_1 = \frac{2\pi}{a}(1/\sqrt{3}, 1)$, $\mathbf{b}_2 = \frac{2\pi}{a}(1/\sqrt{3}, -1)$. A visual representation of the real and reciprocal spaces with these vectors can

be seen in figure 1 below.

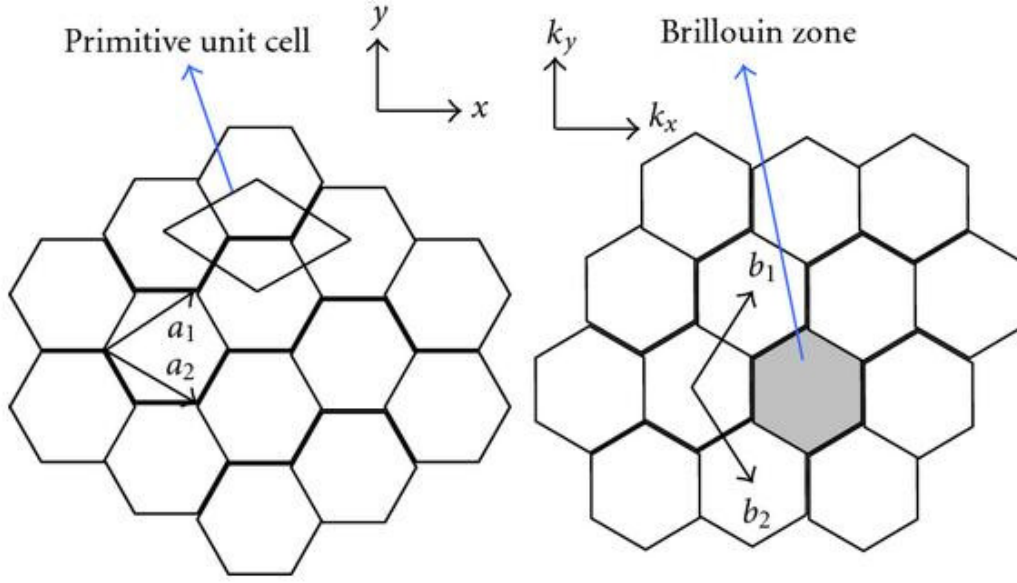


Figure 1: Real and reciprocal space construction for graphene. image taken from [4]

From the figure above we can see that the Brillouin zone is hexagonal, consequently we have that the important symmetry points are the Γ point at the center of the hexagon, M points at the center of the edges and K points at the vertices of the hexagon. K points of symmetry are particularly interesting because the gap between the π energy bands reduces to zero in those, as we will see later. A practical recap of important vectors and points in different coordinates is shown in the table 1 below.

Real space	Cartesian	Relative to base
$\mathbf{a}_1$	$a(\sqrt{3}/2, 1/2)$	$(1, 0)$
$\mathbf{a}_2$	$a(\sqrt{3}/2, -1/2)$	$(0, 1)$
A	$(0, 0)$	$(0, 0)$
B	$(2/\sqrt{3}, 0)$	$(2/3, 2/3)$
Reciprocal space	Cartesian	Relative to base
$\mathbf{b}_1$	$2\pi/a(1/\sqrt{3}, 1)$	$(1, 0)$
$\mathbf{b}_2$	$2\pi/a(1/\sqrt{3}, -1)$	$(0, 1)$
Γ	$(0, 0)$	$(0, 0)$
M	$2\pi/a(1/\sqrt{3}, 0)$	$(1/2, 1/2)$
K	$2\pi/a(1/\sqrt{3}, 1/3)$	$(2/3, 1/3)$

Table 1: Coordinates of graphene crystal vectors and points.

With a few considerations, we can notably simplify the graphene tight-binding model to a form that can be more easily solved:

- The most relevant part of the band structure that we want to focus on is the valence band, which is formed by the π orbitals which are built from the p_z atomic orbitals.
- Since the distance of second nearest neighbour atoms is almost double the distance of nearest neighbours we can take into account only the former, as a reasonable first approximation

- The system is a crystal and in general atoms of type A only have nearest neighbours of type B and viceversa

These translate to:

- We take only a single orbital per atom into account
- We keep only nearest neighbour matrix elements in the tight-binding matrices (see eq. [11] to [15], the overlap integrals between more distant atoms are approximated to zero)
- We can consider a single atom of type A with its 3 neighbours of type B (and viceversa) to encode the solution on the whole crystal

1.4.1 Derivation of the formula 6.20. (See "Computational Physics" by J.M. Thijssen[6])

It's important to note that our specific case is a bit different from the general explanation of section 1.3: instead of multiple orbitals per atom and just one atom (and its equivalents in the crystal), we are dealing with just one orbital per atom but a basis of two non equivalent atoms. Because of this, the equations will be slightly different. We can now more explicitly write eq. [9], [10] and [11] for this approximated system using the considerations above:

$$\phi_{j,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} p_z(\mathbf{r} - \mathbf{R}) \quad (16)$$

These are the Bloch basis functions for all atoms of type j , where $j = A, B$, constructed from p_z orbitals. N is the number of unit cells in the crystal. A general Bloch state is then:

$$\phi_{\mathbf{k}}(\mathbf{r}) = C_A(\mathbf{k})\phi_{A,\mathbf{k}} + C_B(\mathbf{k})\phi_{B,\mathbf{k}} \quad (17)$$

Renaming $C_j(\mathbf{k})$ to $\psi_j(\mathbf{k})$ for consistency with the book by Thijssen, eq. [11] becomes:

$$\begin{pmatrix} H_{AA}(\mathbf{k}) & H_{AB}(\mathbf{k}) \\ H_{BA}(\mathbf{k}) & H_{BB}(\mathbf{k}) \end{pmatrix} \begin{pmatrix} \psi_A(\mathbf{k}) \\ \psi_B(\mathbf{k}) \end{pmatrix} = E(\mathbf{k}) \begin{pmatrix} S_{AA}(\mathbf{k}) & S_{AB}(\mathbf{k}) \\ S_{BA}(\mathbf{k}) & S_{BB}(\mathbf{k}) \end{pmatrix} \begin{pmatrix} \psi_A(\mathbf{k}) \\ \psi_B(\mathbf{k}) \end{pmatrix} \quad (18)$$

The terms $H_{AA}(\mathbf{k}) = H_{BB}(\mathbf{k})$, $S_{AA}(\mathbf{k}) = S_{BB}(\mathbf{k})$, $H_{BA}(\mathbf{k}) = H_{AB}^*(\mathbf{k})$, $S_{BA}(\mathbf{k}) = S_{AB}^*(\mathbf{k})$ and will have to be calculated for each $\mathbf{k}$. Assuming to know their values, the eigenvalue problem above gives the spectrum $E(\mathbf{k})$. Omitting the explicit $\mathbf{k}$ dependency for simplicity we can rewrite it to:

$$\begin{pmatrix} H_{AA} - ES_{AA} & H_{AB} - ES_{AB} \\ H_{AB}^* - ES_{AB}^* & H_{AA} - ES_{AA} \end{pmatrix} \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = 0 \quad (19)$$

For this problem to have nontrivial solutions the matrix A has to be non invertible, thus by setting $\det(A) = 0$ we get:

$$H_{AA}^2 - 2EH_{AA}S_{AA} + E^2S_{AA}^2 - H_{AB}H_{AB}^* + EH_{AB}S_{AB}^* + EH_{AB}^*S_{AB} - E^2S_{AB}S_{AB}^* = 0 \quad (20)$$

Rearranging terms and then renaming them to make the equation more clear:

$$E^2(S_{AA}^2 - S_{AB}S_{AB}^*) + E(H_{AB}S_{AB}^* + H_{AB}^*S_{AB} - 2H_{AA}S_{AA}) + (H_{AA}^2 - H_{AB}H_{AB}^*) = 0 \quad (21)$$

We can then simply apply the quadratic equation to get:

$$E_{\pm}(\mathbf{k}) = \frac{-(-2E_0 + E_1) \pm \sqrt{(-2E_0 + E_1)^2 - 4E_2E_3}}{2E_3} \quad (22)$$

Where

$$\begin{aligned} E_0 &= H_{AA}S_{AA} & E_1 &= H_{AB}S_{AB}^* + H_{AB}^*S_{AB} \\ E_2 &= H_{AA}^2 - H_{AB}H_{AB}^* & E_3 &= S_{AA}^2 - S_{AB}S_{AB}^* \end{aligned}$$

1.4.2 Derivation of formulas 6.21, 6.22 and 6.23 (See "Computational Physics" by J.M. Thijssen[6])

We have derived the solution for the energy spectrum eigenvalues, but to use it we have to know the matrix elements for each $\mathbf{k}$: let's calculate them in the case of nearest neighbour approximation.

1. The term S_{AA} is simply 1, since we are assuming the basis functions to be normalized.
2. The term $H_{AA} = \frac{1}{N} \sum_{\mathbf{R}} \sum_{\mathbf{R}'} e^{i\mathbf{k}(\mathbf{R}_A - \mathbf{R}'_A)} \langle p_z(\mathbf{r} - \mathbf{R}'_A) | H | p_z(\mathbf{r} - \mathbf{R}_A) \rangle$

for nearest neighbours becomes

$$H_{AA} = \frac{1}{N} \sum_{\mathbf{R}_A} \langle p_z(\mathbf{r} - \mathbf{R}_A) | H | p_z(\mathbf{r} - \mathbf{R}_A) \rangle = \langle p_z(\mathbf{r} - \mathbf{R}_A) | H | p_z(\mathbf{r} - \mathbf{R}_A) \rangle = \epsilon_{2p}$$

Because of crystal symmetry. ϵ_{2p} is a constant independent of $\mathbf{k}$ that expresses the interaction of the orbitals with themselves (other orbitals of type A are at least second nearest neighbours so they are ignored).

3. The term $S_{AB} = \frac{1}{N} \sum_{\mathbf{R}_B} \sum_{\mathbf{R}_A} e^{i\mathbf{k}(\mathbf{R}_B - \mathbf{R}_A)} \langle p_z(\mathbf{r} - \mathbf{R}_A) | p_z(\mathbf{r} - \mathbf{R}_B) \rangle$

by taking atoms of type A and counting only the nearest neighbours, which are 3 atoms of type B, we can write the position of the first B atom $\mathbf{R}_{B1}$ as $\mathbf{R}_A + \mathbf{R}_{11}$ and likewise for the other atoms to get:

$$S_{AB} = \frac{1}{N} \sum_{\mathbf{R}_A} \sum_{j=1}^3 e^{i\mathbf{k} \cdot \mathbf{R}_{1j}} \langle p_z(\mathbf{r} - \mathbf{R}_A) | p_z(\mathbf{r} - \mathbf{R}_A - \mathbf{R}_{1j}) \rangle = s_0 (e^{i\mathbf{k} \cdot \mathbf{R}_{11}} + e^{i\mathbf{k} \cdot \mathbf{R}_{12}} + e^{i\mathbf{k} \cdot \mathbf{R}_{13}})$$

where $s_0 = \langle p_z(\mathbf{r} - \mathbf{R}_A) | p_z(\mathbf{r} - \mathbf{R}_A - \mathbf{R}_{1j}) \rangle$ with any j because of symmetry.

4. The term $H_{AB} = \frac{1}{N} \sum_{\mathbf{R}_B} \sum_{\mathbf{R}_A} e^{i\mathbf{k}(\mathbf{R}_B - \mathbf{R}_A)} \langle p_z(\mathbf{r} - \mathbf{R}_A) | H | p_z(\mathbf{r} - \mathbf{R}_B) \rangle$, with the same treatment as S_{AB} becomes

$$\gamma_0 (e^{i\mathbf{k} \cdot \mathbf{R}_{11}} + e^{i\mathbf{k} \cdot \mathbf{R}_{12}} + e^{i\mathbf{k} \cdot \mathbf{R}_{13}})$$

where

$$\gamma_0 = \langle p_z(\mathbf{r} - \mathbf{R}_A) | H | p_z(\mathbf{r} - \mathbf{R}_A - \mathbf{R}_{1j}) \rangle$$

Going back to eq. [22], we can now rewrite the terms E_0 to E_3 :

$$E_0 = \epsilon_{2p}$$

$$E_1 = \gamma_0 s_0 [6 + 2 \sum_{j \neq k} (e^{i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k})} + e^{i\mathbf{k} \cdot (\mathbf{R}_{1k} - \mathbf{R}_{1j})})] = 2\gamma_0 s_0 (3 + 2 \sum_{k > j} \cos(i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k})))$$

$$E_2 = \epsilon_{2p}^2 - \gamma_0^2[3 + \sum_{j \neq k} (e^{i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k})} + e^{i\mathbf{k} \cdot (\mathbf{R}_{1k} - \mathbf{R}_{1j})})] = \epsilon_{2p}^2 - \gamma_0^2[3 + 2 \sum_{k > j} \cos(i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k}))]$$

$$E_3 = 1 - s_0^2[3 + \sum_{j \neq k} (e^{i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k})} + e^{i\mathbf{k} \cdot (\mathbf{R}_{1k} - \mathbf{R}_{1j})})] = 1 - s_0^2[3 + 2 \sum_{k > j} \cos(i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k}))]$$

where the sums more explicitly are:

$$\sum_{k > j} \cos(i\mathbf{k} \cdot (\mathbf{R}_{1j} - \mathbf{R}_{1k})) = \cos(i\mathbf{k} \cdot (\mathbf{R}_{11} - \mathbf{R}_{12})) + \cos(i\mathbf{k} \cdot (\mathbf{R}_{11} - \mathbf{R}_{13})) + \cos(i\mathbf{k} \cdot (\mathbf{R}_{12} - \mathbf{R}_{13}))$$

Now, if we look at the crystal structure of graphene in figure 1 we can see that we can take any two vector differences as primitive lattice vectors, for example $\mathbf{R}_{11} - \mathbf{R}_{12} = \mathbf{a}_1$; $\mathbf{R}_{11} - \mathbf{R}_{13} = \mathbf{a}_2$

Consequently we can define the function

$$f(\mathbf{k}) = 3 + 2 \cos(\mathbf{k} \cdot \mathbf{a}_1) + 2 \cos(\mathbf{k} \cdot \mathbf{a}_2) + 2 \cos(\mathbf{k} \cdot (\mathbf{a}_1 - \mathbf{a}_2)) \quad (23)$$

If we decompose $\mathbf{k}$ in the reciprocal space basis $\mathbf{k} = k_{b1}\mathbf{b}_1 + k_{b2}\mathbf{b}_2$ we get:

$$f(\mathbf{k}) = 3 + 2 \cos(2\pi k_{b1}) + 2 \cos(2\pi k_{b2}) + 2 \cos(2\pi(k_{b1} - k_{b2})) \quad (24)$$

Or, in cartesian coordinates:

$$f(\mathbf{k}) = 3 + 2 \cos(a(\sqrt{3}/2k_x + 1/2k_y)) + 2 \cos(a(\sqrt{3}/2k_x - 1/2k_y)) + 2 \cos(ak_y) \quad (25)$$

Inserting f and the previous results into eq. [22], we get:

$$E_{\pm}(\mathbf{k}) = \frac{\epsilon_{2p} \mp \gamma_0 \sqrt{f(\mathbf{k})}}{1 \mp s_0 \sqrt{f(\mathbf{k})}} \quad (26)$$

An important consequence of this is that the bands touch in the points where $f(\mathbf{k}) = 0$: if we substitute the coordinates of Brillouin zone K points from table 1 in eq. [24] or [25] we can see that they satisfy this condition.

2 Single-walled carbon nanotubes

Single-walled carbon nanotubes (SWCNTs), from now on simply referred to as "nanotubes", are graphene sheets rolled into a cylindrical form. This means that one direction of the plane becomes finite (the circumference of the nanotube) and another perpendicular to it remains (theoretically) infinite and indicates the longitudinal direction of the nanotube. We can thus imagine obtaining a nanotube by running two parallel cuts through a graphene sheet and then rolling and gluing together the sides of the removed piece. There exist different ways of "cutting" and rolling up a graphene plane and thus different types of nanotubes, based on their structure.

Mathematically, we can define the different types of nanotubes by the direction and length of the vector that separates the two cuts on the graphene sheet. To maintain a crystalline structure, these vectors must be restricted to integer linear combinations of the graphene primitive lattice vectors, so that by going around the circumference we start and end in equivalent crystalline positions (for example, start and end on an atom of type A, see figure 2). We can then identify a type of nanotube with the integer factors (n, m) , saying that its rolled up circumference corresponds to the $\mathbf{c}_h = n\mathbf{a}_1 + m\mathbf{a}_2$ vector of length L on the graphene sheet, often named "chiral" vector. $(n, 0)$ nanotubes are called "zigzag" tubes because the circumferences are made up of zigzag structures of nearest neighbour bonds, while (n, n) nanotubes are called "armchair" nanotubes. In the band structure calculations we neglect the effects of the curvature of the tube, which is an approximation that fits better with large n and/or m .

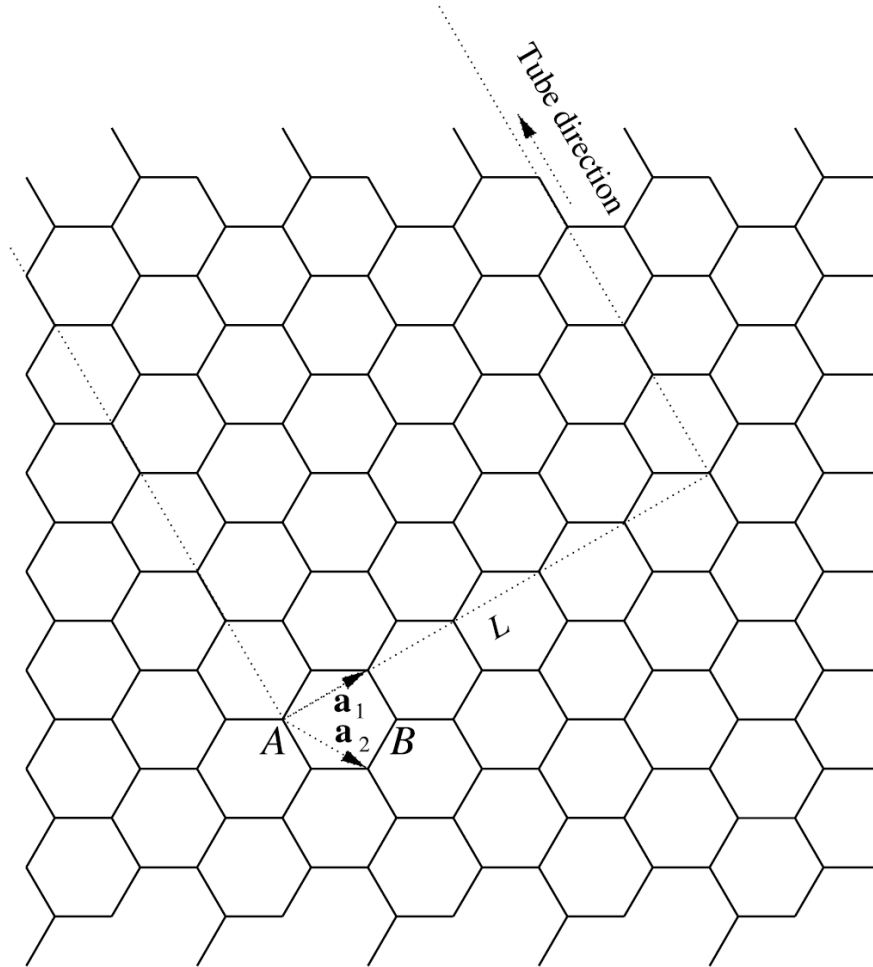


Figure 2: Visual representation of nanotube cutting from a graphene sheet. This specific case represents the "zigzag" $(5,0)$ nanotube. Image courtesy of the book by Thijssen [6].

An important thing to note is that in common with graphene (and in general with crystals) the electronic density and the potential are periodic, but in addition to this in nanotubes the wave functions themselves become strictly periodic across the tube. This is because by doing a revolution along the tube we get back to the same point: mathematically this condition is expressed by $\mathbf{k} \cdot \mathbf{c}_h = 2\pi j$ where j is an integer and has some parallels with periodic boundary conditions for crystals. Let's use this condition and see its consequences:

- We define the rectangular unit cell of the nanotube by the chiral vector $\mathbf{c}_h$ and the transverse vector $\mathbf{T}$, and the reciprocal unit cell with the corresponding $\mathbf{b}_\perp$ and $\mathbf{b}_\parallel$
- $k_\perp = \mathbf{k} \cdot \frac{\mathbf{c}_h}{|\mathbf{c}_h|} = \frac{2\pi j}{L}$ shows that the transverse $k_\perp$ products of $\mathbf{k}$ are discretized.
- Equivalently, defining $\mathbf{k}_\perp == k_{b_\perp} \mathbf{b}_\perp$ and using $\mathbf{b}_\perp \cdot \mathbf{c}_h = 2\pi$ we find that in general $\mathbf{k}_\perp = j\mathbf{b}_\perp$.
- The states are then labelled by longitudinal $\mathbf{k}_\parallel == k_{b_\parallel} \mathbf{b}_\parallel$ vectors along the tube and for each fixed longitudinal $\mathbf{k}_\parallel$ vector we find a discrete energy spectrum.
- The brillouin zone becomes one dimensional and goes from the point Γ to a point labelled X .

In practice the nanotube brillouin zone can be visualized as a series of one dimensional lines that pass through the 2D brillouin zone of graphene, and the carbon nanotube will be of metallic nature when the discretization of $\mathbf{k}_\perp$ allows one or more lines to pass by graphene K points in reciprocal space. Since the sliced unit cell is in general rectangular, we have for the nanotube reciprocal lattice vectors $\mathbf{b}_\perp \parallel \mathbf{c}_h$; $\mathbf{b}_\parallel \parallel \mathbf{T}$.

2.1 (n,0) zigzag nanotubes

Let's consider the case of a $(n,0)$ zigzag tube: in this type of system, as we can see from figure 2, the longitudinal period $|\mathbf{T}|$ (distance required to end up in an equivalent point) is $a\sqrt{3}$ where a is the length of the basis vectors, so the brillouin zone runs up to $k_\parallel = \mathbf{k} \cdot \frac{\mathbf{T}}{|\mathbf{T}|} = \pi/a\sqrt{3}$. The transverse period is given by the circumference $L = na$ and the possible transverse $k_\perp$ are $2\pi j/na$, with j going from 1 to n (higher or lower integers represent equivalent states because of periodicity). Since in this case $\mathbf{b}_\perp = \frac{2\pi}{na^2}\mathbf{a}_1$, looking at the coordinates of table 1, we see that $\mathbf{k}_\perp = j\mathbf{b}_\perp$ certainly touches the K point when $j/n = 2/3$ (or equivalently $k_\perp = 2\pi j/na = 4\pi/3a$), which can happen for every n that is a multiple of 3.

2.2 (n,n) armchair nanotubes

In the case of an (n,n) armchair tube, in the graphene sheet represented in figure 2 we would be performing two straight vertical cuts. It can be seen then that the longitudinal period would be a while the transverse period would be $na\sqrt{3}$, so the brillouin zone runs up to $k_\parallel = \pi/a$ (equivalently $k_{b_\parallel}$ up to $1/2$) and the possible transverse $k_\perp$ are $2\pi j/na\sqrt{3}$. Since in this case $\mathbf{k}_x \parallel \mathbf{b}_\perp$; $\mathbf{k}_y \parallel \mathbf{b}_\parallel$, $|\mathbf{b}_\perp| = \frac{2\pi}{na\sqrt{3}}$; $|\mathbf{b}_\parallel| = \frac{2\pi}{a}$ looking at the cartesian coordinates of K in table 1 we can see that the condition for reaching it becomes $j/n = 1$ (or equivalently $2\pi j/na\sqrt{3} = 2\pi/a\sqrt{3}$) which can be satisfied for each n , so (n,n) nanotubes have always a zero bandgap.

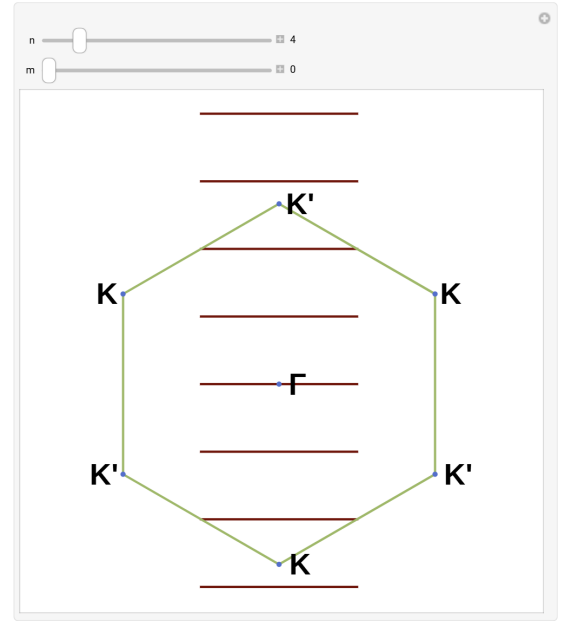
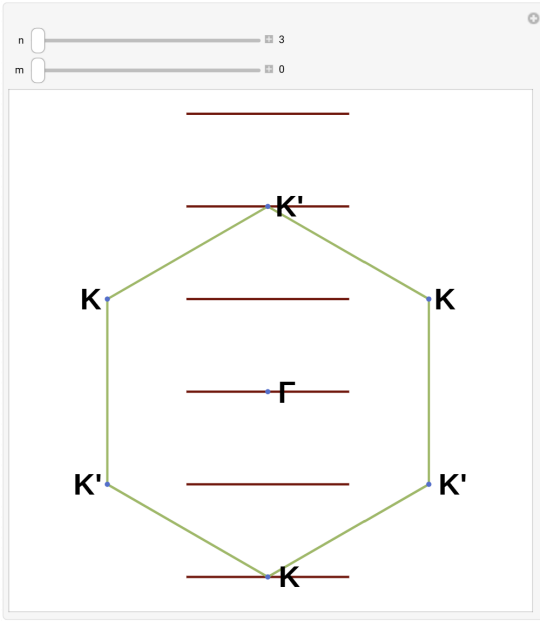


Figure 3: Brillouin zones of respectively $(3,0)$ and $(4,0)$ nanotubes superimposed on the one of graphene, plotted using the website [2]

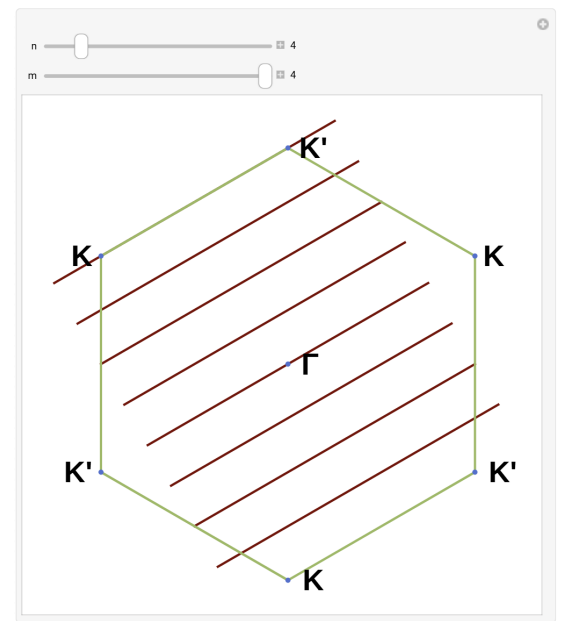
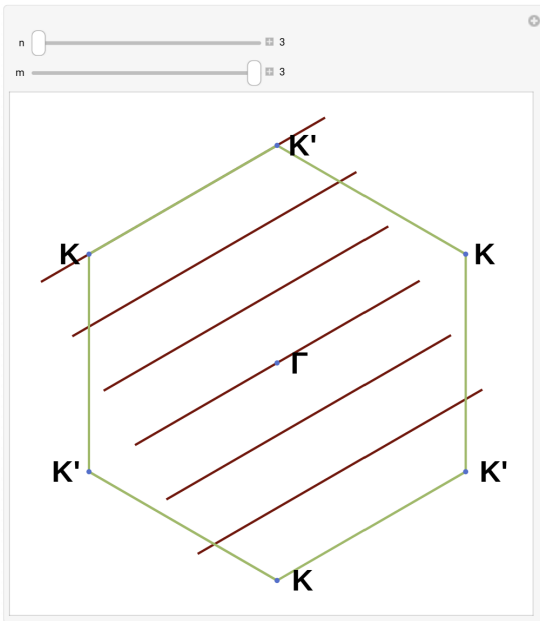


Figure 4: Brillouin zones of respectively $(3,3)$ and $(4,4)$ nanotubes superimposed on the one of graphene, plotted using the website [2]

2.3 Generic (n,m) chiral nanotubes

In the more general (n, m) case often called "chiral" nanotube, it can be seen by geometric properties (law of cosines) that $L = |\mathbf{c}_h| = a\sqrt{n^2 + m^2 + nm}$. If we introduce the primitive longitudinal translation vector and express it with the graphene primitive lattice vectors $\mathbf{T} = t_1\mathbf{a}_1 + t_2\mathbf{a}_2$, by using the property $\mathbf{T} \perp \mathbf{c}_h$ we can demonstrate that $t_2 = -\frac{2n+m}{2m+n}t_1$ and so t_1 and t_2 will have the general form:

$$t_1 = \frac{n+2m}{d_R}; \quad t_2 = -\frac{2n+m}{d_R} \quad (27)$$

Since $\mathbf{C}_h$ and $\mathbf{T}$ represent the unit cell of the nanotube which is made from a graphene crystal, t_1 and t_2 have to be the lowest possible integer numbers for a fixed (n, m) pair thus d_R is the highest common denominator of $n+2m$ and $2n+m$ which if it's a multiple of 3 can be expressed like $3d$ without loss of generality. This happens when $n-m$ is a multiple of 3 (subtract the first term to the second to see it) and it's also the general condition for a carbon nanotube to have a band gap of zero, the previously explained conditions for $(n, 0)$ and (n, n) nanotubes were special cases of this rule.

When trying to compute the energy bands of (n, m) chiral nanotubes we have to find the allowed $\mathbf{k}$ ranges in reciprocal space and be able to express them with respect to k_x, k_y cartesian or relative k_{b1}, k_{b2} coordinates so we can use the expressions from eq. [24] or [25] and plug them into eq. [26]: after all, following our discourse so far the energy spectrum of carbon nanotubes maintains the same expression of the one of graphene, just restricted to specific one-dimensional brillouin zones. Let's define:

$$q = 2(n^2 + m^2 + nm)/d_R = 2L^2/d_R a^2 \quad (28)$$

The Γ points $\mathbf{k}_\Gamma(j) = j\mathbf{b}_\perp$ at the center of each of the nanotube reciprocal space lines can be expressed in terms of graphene reciprocal vectors. By using $\mathbf{b}_\perp = k_{b1\perp}\mathbf{b}_1 + k_{b2\perp}\mathbf{b}_2$, $\mathbf{b}_\perp \perp \mathbf{T}$, $\mathbf{b}_\perp \cdot \mathbf{c}_h = 2\pi$ we can show that:

$$\mathbf{k}_\Gamma(j) = j \left(\frac{2n+m}{qd_R}\mathbf{b}_1 + \frac{n+2m}{qd_R}\mathbf{b}_2 \right) \quad (29)$$

for a (n, m) nanotube where d_R is the greatest common denominator of $n-m$ as before and j is an integer that goes from $-q/2$ to $q/2 - 1$. The components of the vector $\mathbf{b}_\parallel = k_{b1\parallel}\mathbf{b}_1 + k_{b2\parallel}\mathbf{b}_2$ can be determined using $\mathbf{b}_\parallel \perp \mathbf{c}_h$, $\mathbf{b}_\parallel \cdot \mathbf{T} = 2\pi$. The endpoints of each line, that correspond to X and equivalent points in the 1-D brillouin zone that were mentioned before, are $\mathbf{k}_\Gamma(j) \pm \mathbf{k}_z$ where:

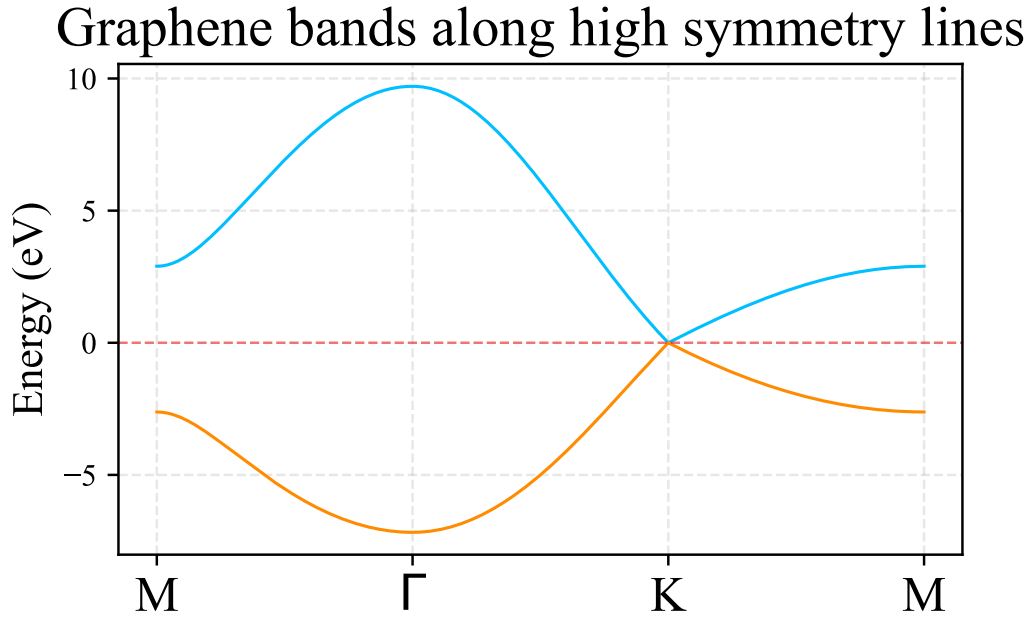
$$\mathbf{k}_z = \frac{\mathbf{b}_\parallel}{2} = \frac{m}{2q}\mathbf{b}_1 - \frac{n}{2q}\mathbf{b}_2 \quad (30)$$

In conclusion, to obtaining the energy bands in the one dimensional brillouin zone (between Γ and X for each j) for a (n, m) nanotube can be done by calculating each $E(\mathbf{k}_\Gamma)$ to $E(\mathbf{k}_\Gamma + \mathbf{k}_z)$ for graphene in relative coordinates using eq. [24], [26], [28], [29] and [30].

3 Computation of band structure of graphene and carbon nanotubes

The program that performs the tight-binding band structure simulations has been written in Python and is publicly available on GitHub at the following link: [TB Graphene Nanotubes](#) (also cited in [1]). Please refer to its README.md for a more detailed explanation on how it's structured and how to use it.

Below, I will give some examples of its outputs and their comparisons with band structure images taken from the book by Thijssen [6] and the paper [5].



(a) Graphene nearest neighbour tight-binding π band dispersion calculated with [1] (precision parameter = 100). Estimated bandgap: 0.000 eV

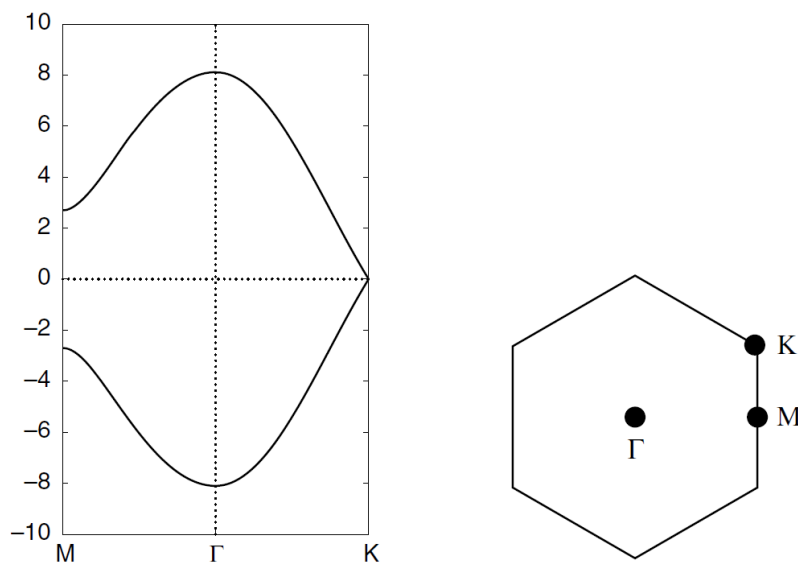


Figure 6: Tight-binding band structure of graphene, the points of the Brillouin zone are shown on the right hand side. Figure 6.5. in the book by Thijssen [6]

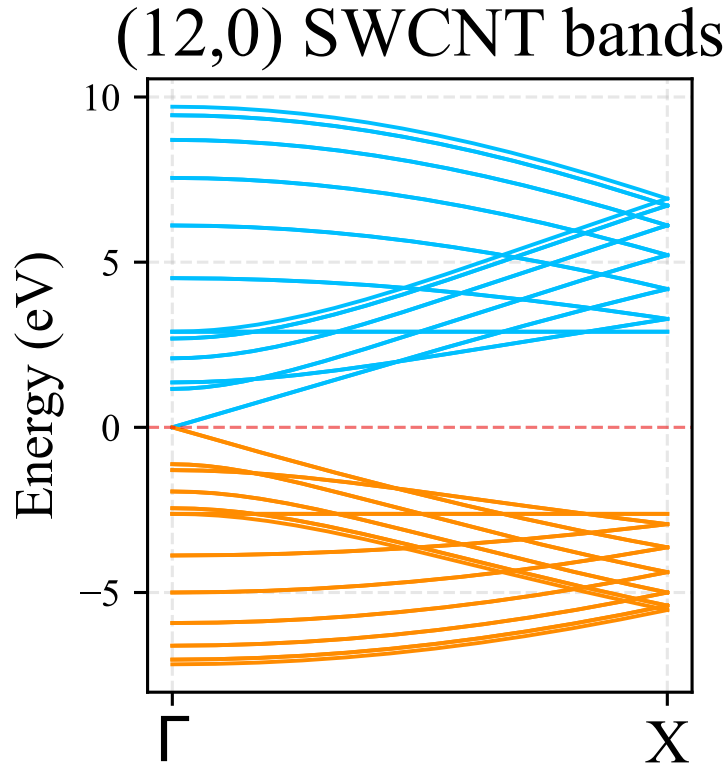


Figure 7: (12,0) nanotube nearest neighbour tight-binding band dispersion calculated with [1] (precision parameter = 100). Estimated bandgap: 0.000 eV

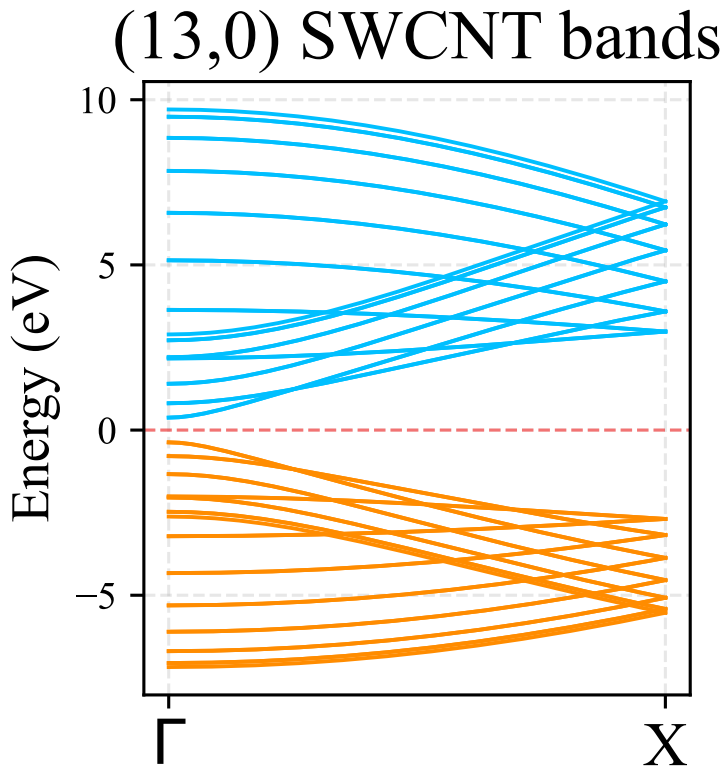


Figure 8: (13,0) nanotube nearest neighbour tight-binding band dispersion calculated with [1] (precision parameter = 100). Estimated bandgap: 0.749 eV

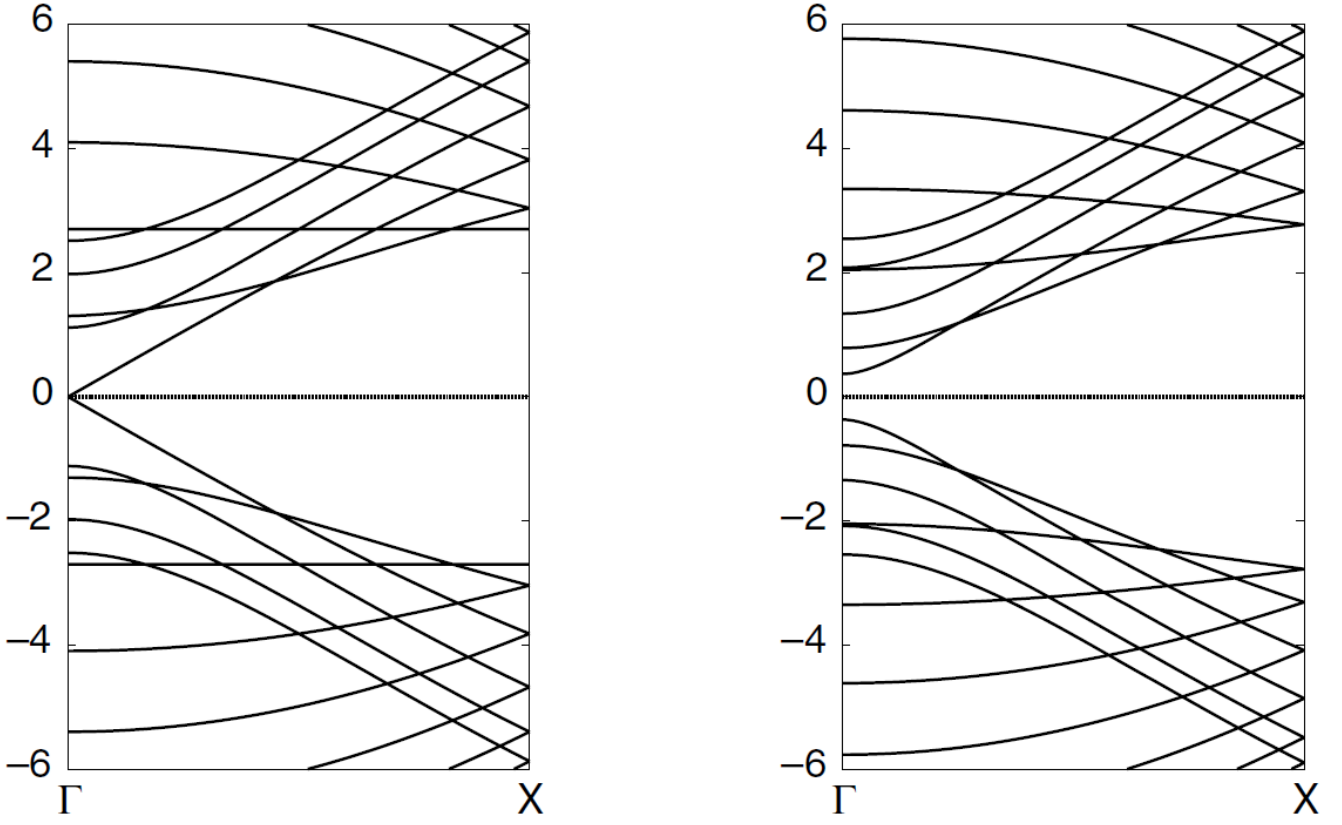


Figure 9: Energy bands for a (12, 0) and a (13,0) tube: in reality, both have an energy gap. That of the even tube is very small and it is absent in the present analysis owing to the neglect of tube curvature. Figure 6.6. in the book by Thijssen [6].

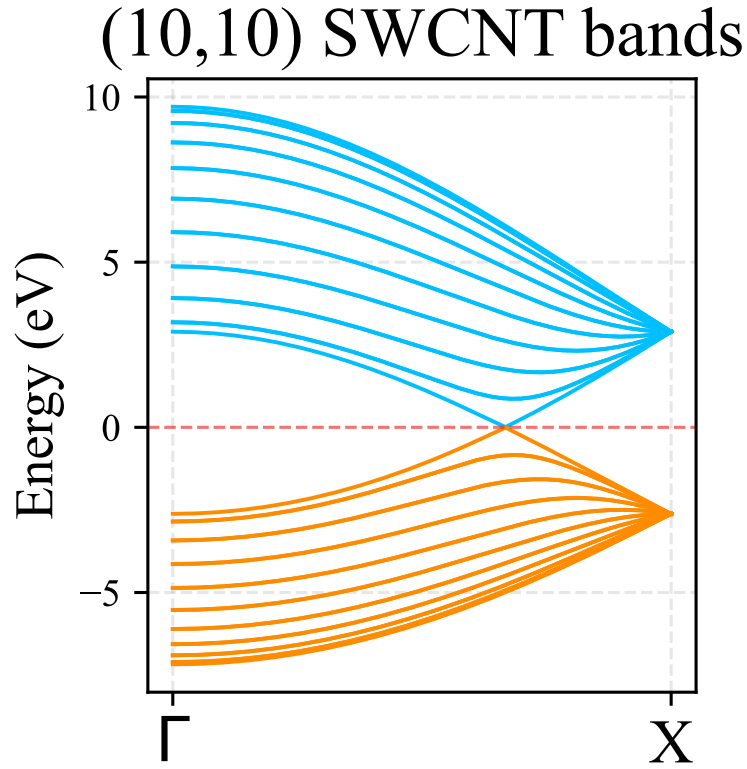


Figure 10: (10,10) armchair nanotube nearest neighbour tight-binding band dispersion calculated with [1] (precision parameter = 100). Estimated bandgap: 0.000 eV

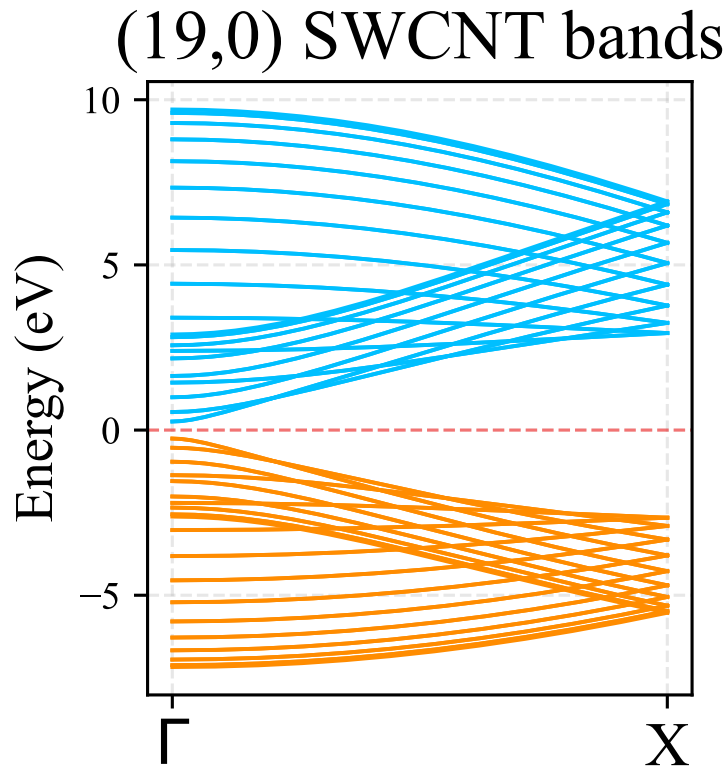


Figure 11: (19,0) zigzag nanotube nearest neighbour tight-binding band dispersion calculated with [1] (precision parameter = 100). Estimated bandgap: 0.516 eV

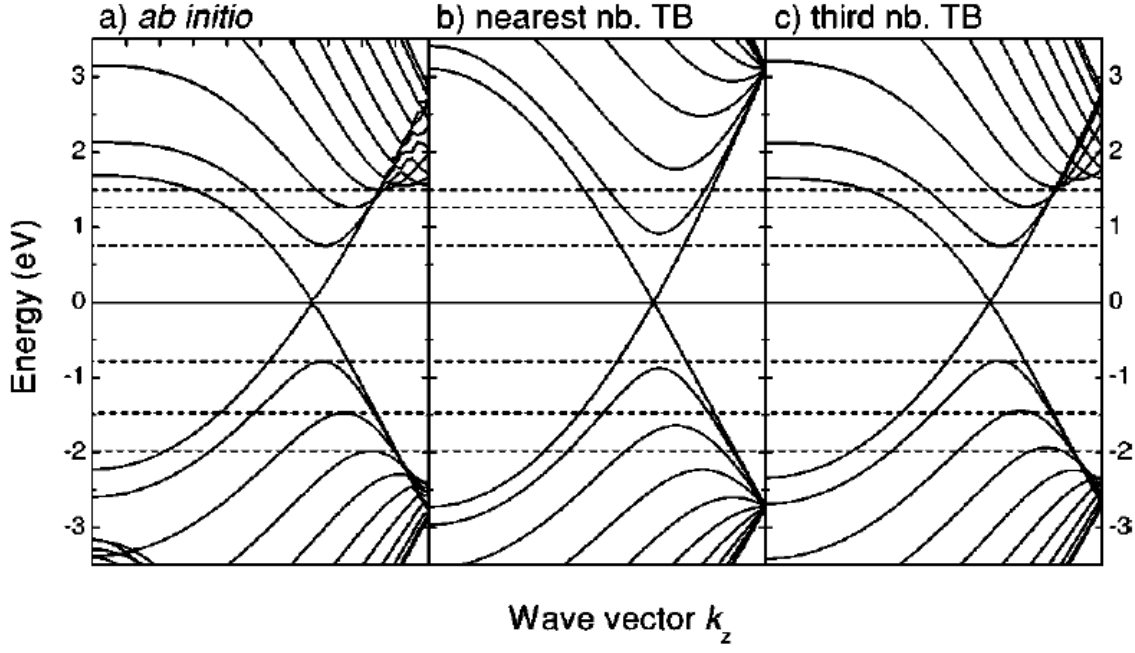


Figure 12: Band structure of a (10,10) armchair nanotube calculated in different ways. The important one in our context is (b) Nearest-neighbor tight-binding calculation with parameters obtained from a fit to the optical energy range. The dashed lines denote ab initio calculated energies of the singularities in the density of states. Taken from paper [5] (Fig.4)

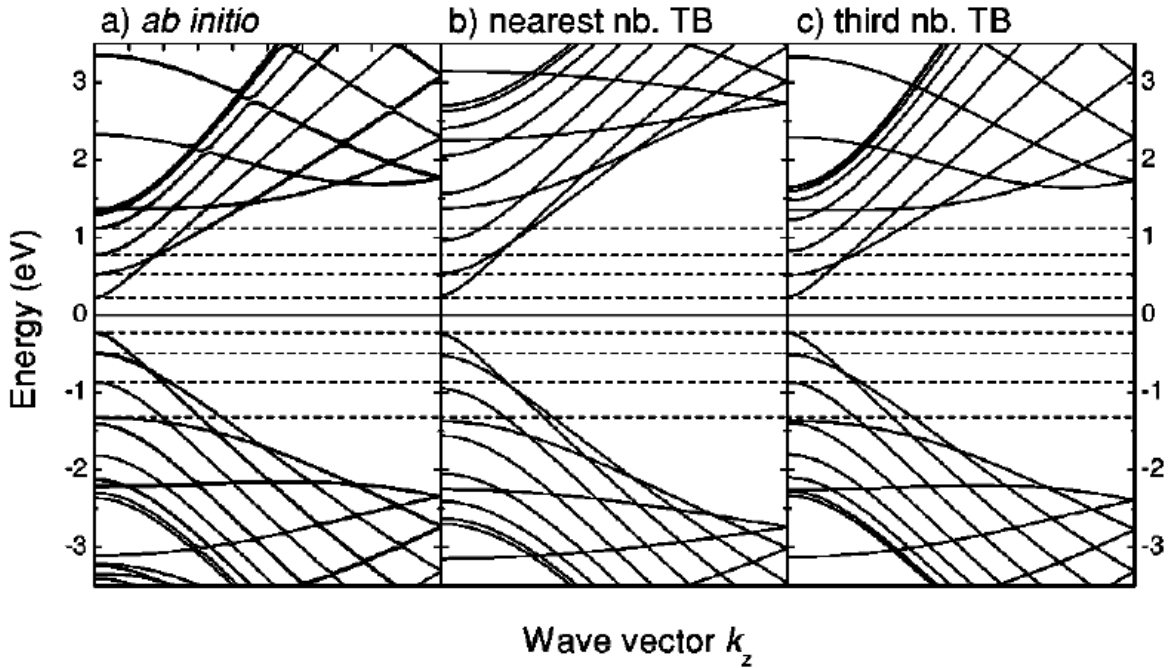


Figure 13: Band structure of a (19,0) zigzag nanotube calculated in different ways. The important one in our context is (b) Nearest-neighbor tight-binding calculation with parameters obtained from a fit to the optical energy range. The dashed lines denote ab initio calculated energies of the singularities in the density of states. Taken from paper [5] (Fig.5)

From the images above we can see that the Python program was successful at replicating the reference results, with the visual and small numeric differences explainable by the slightly different choices of tight-binding parameters and plotting options.

References

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